

The Learning Loop That Learns Itself

From Recursive Self-Reference to Phase-Compatible Self-World Reinstatement in Self-Aware Networks

Micah Blumberg

Self Aware Networks Research Institute

selfawarenetworks.com

Preprint - August 13, 2026

Abstract

Self-reference, learned self-models, and conscious self-awareness are distinct problems. This paper reconstructs a dated Self Aware Networks (SAN) proposal that joins them through an embodied learning loop: sensory and internally generated states contribute to action; typed bodily and worldly consequences return; receiver and route dispositions change; and later phase-compatible activity can reinstate and revise a distributed self-world state. Hofstadter's strange loop and Hawkins's predictive learning supply different conceptual donors. The explicit phrase "a learning loop that learns itself" appears in a September 2011 SAN source; PWD, the Gamma Consideration Sandwich (GCS), NAPOT, and Artificial Neurology are later extensions and are not backdated.

The paper combines source genealogy, 37 external/conceptual references, nine SAN sources, a typed causal model, 12 countermodels, a deterministic reference application, two frozen synthetic development studies, and bounded Lean verification. Development Study 1 completed 88 runs and 9,152 episodes but passed only two of six gates; full versus yoked return was an exact null. A separately frozen shortcut-resistant study completed 176 runs and 28,160 episodes, passing five of seven gates. It identified contingent return, persistent transfer, exact remapping, and receiver-relative input use, but phase scrambling did not impair partial-cue success and a matched generic recurrent learner tied the full implementation exactly. All four canonical files replayed byte-for-byte. Lean checked 33 finite interface, mapping, result-accounting, custody, and claim-boundary theorems with exit code 0 and no admitted proofs.

The resulting claim is bounded: the package realizes and partially identifies a finite learning-loop operator. It does not validate biological SAN, consciousness, the complete GCS mapping, phase necessity, or SAN-specific algorithmic superiority. The adverse results define the next empirical work rather than being removed from the theory.

1. The problem: four operations, not one

Self-reference, self-modeling, learning-loop closure, and conscious self-world rendering are often compressed into one idea. They are different operations.

First, a symbol or process can refer to itself. Hofstadter's strange-loop framework describes recursive level crossing in which the system's representational activity comes to include an "I" that appears to act back on the process that constructs it [1,2]. Hofstadter uses ordinary feedback, including audio and video feedback, to motivate looping before distinguishing the abstract symbol-level loop of selfhood. This contribution is conceptual: it does not specify neural phase coherence, a PWD operator, or a sensorimotor learning rule.

Second, a nervous system can learn a model that includes the body, perspective, action repertoire, memory, expectation, and social identity. Hawkins and Blakeslee emphasized prediction and recurring cortical operations [3]. Predictive-coding models likewise show how hierarchical feedback can encode expectations while residual signals carry mismatch information [5]. Multisensory manipulations demonstrate that bodily self-attribution is plastic: synchronous visual, tactile, and proprioceptive relations can shift experienced hand ownership [6], and focal electrical stimulation can disturb whole-body perspective and location [7]. These findings support learned and constructed body-related representation; they do not identify the whole self with one model.

Third, a learning system can be closed through action. Its present state contributes to an action; the action changes the body and world; those changes return through sensory channels; and the return alters later processing. In this operation, the system repeatedly encounters evidence partly caused by its own preceding state. This is stronger than recurrence alone because the environment and body participate in the state transition.

Fourth, SAN proposes that the distributed network state is not merely a message sent to an unspecified inner witness. The same embodied network receives, transforms, projects, and recurrently reconstructs the changing pattern. A self-model is one learned content within that process. The term *self* refers to the whole active learning loop whose contents, dispositions, and consequences remain mutually causal.

The paper's thesis is therefore not “a self is a feedback loop,” “prediction is consciousness,” or “gamma is the seat of awareness.” It is that a biological self can be modeled as a nested embodied learning loop whose own action-conditioned consequences update the learned network that will construct the next self-world state. SAN's later phase-differential vocabulary proposes one candidate mechanism by which that reconstruction becomes receiver-specific, temporally selective, differentiated, and testable.

2. Dated SAN genealogy

The genealogy must preserve the difference between intellectual influence, dated source text, later mechanistic development, and current synthesis.

2.1 Influence stage: 1999 and 2005

Blumberg reports first reading Hofstadter in 1999 and initially framing the problem as a loop aware of itself. After reading *On Intelligence* in 2005, he reconceptualized the problem as a loop capable of learning itself. These are author-reported influence dates recorded retrospectively in 2026, not immutable 1999 or 2005 public custody records. Hofstadter and Hawkins remain conceptual donors; neither is retroactively assigned SAN's later biological mechanisms.

2.2 September 2011: the explicit learning loop

A source dated September 21, 2011 and publicly fixed in Git in 2022 gives the first exact formulation reviewed for this paper [38]. It describes sensation changing neural activity, neural activity contributing to muscle action, action returning through the senses, and the resulting feedback loop learning. It then names the self “a learning loop that learns itself.” A companion source assembled from the same pre-September-21 dialogue makes the three-part structure explicit-senses, brainwaves, and actions-and states that “you learn as you loop” [39].

The strongest source-faithful interpretation is not that a detached symbol has learned a complete inventory of its synapses. The whole active process learns regularities involving its sensations, predictions, actions, and returned consequences. The self-related representations are contents inside that process; the “self” in the formulation is the process as a whole.

2.3 April 2012: the network both plays and listens

An April 19, 2012 note describes perception as learned memory-prediction and then asks how the cellular system can be both the producer and receiver of the ongoing activity [40]. Its later author commentary identifies a then-missing bridge: a memory-prediction must be rendered so that a subsequent array or layer can read and interpret it. The note's enduring contribution is therefore not the claim that prediction exhausts every perceptual operation. It is the movement from a learning loop toward a distributed player-listener architecture.

2.4 September 2012: loops inside loops

A September 18, 2012 source describes the person as “a cooperative rivalry of learning loops inside learning loops,” with expectation matching, prediction error, competition, and dissipative continuation [41]. The nested formulation matters because no single sensory, mnemonic, motor, or interoceptive cycle is sufficient. The larger self-world process depends on coordinated subloops operating at different scales and times.

2.5 The phase and route bridge

An undated note publicly fixed in 2022 supplies a direct bridge toward later phase-compatible reinstatement [42]. It distinguishes mere sensing and transmission from learned reception, discrimination, memory, action, and consequence; describes consciousness as an internal reading-and-writing process; invokes phase coincidence, inhibitory coincidence, alternating decoherence, learned routes, associative playback, and attractor-like completion; and describes a “learning playback loop” that models self, environment, and action paths. The note also contains unrelated speculative cosmological material. That material is neither required by nor imported into this paper. Only the bounded neural learning-loop operations are used.

2.6 Artificial Neurology and two timescales

By 2022, the Neo Mind Cycle and Artificial Neurology notes described a substrate writing and reading its own perception-action-memory-prediction cycle [43]. A separate state-and-stage note distinguishes fast changes in active phase or frequency from slower changes in connection, threshold, memory, and structure [44]. This creates a testable engineering requirement: an artificial learning loop must couple a momentary self-world state to retained developmental change and must show that specific experience caused the retained change.

2.7 Gamma Consideration Sandwich

A mixed-authorship 2024 note places gamma in the functional middle between sensory-associated alpha activity and prefrontal-associated beta activity, and assigns gamma a proposed consideration and proprioceptive-coordination role [45]. The adjacent model-generated replies are discovery context, not Blumberg-authored historical evidence. A 2026 author clarification supplies the mature mapping used here: Layer 4 alpha for primary-sensory-side input, Layer 4 beta for prefrontal thought and top-down input, Layer 2/3 gamma for consideration and proprioceptive coordination, and Layer 5 theta/gamma for action-side output and body-derived return [46]. The mature mapping must not be backdated to 2011, 2012, or even the less-specific 2024 note.

2.8 Current joined formulation

The current paper joins the source stages without collapsing them:

recursive self-reference → learned prediction and self/world models → embodied action-return closure → learned receiver structure → tonic context and receiver-relative PWD → phase-compatible route recruitment → differentiated distributed reinstatement → action → returned evidence → fast correction and slow plastic change.

3. What current neuroscience establishes-and what it does not

No single cited study tests this complete chain. The evidence is organized by component so that support for one edge cannot be passed off as proof of the whole architecture.

3.1 Prediction and mismatch are learned relations

Rao and Ballard showed that a hierarchical predictive-coding model could account for extra-classical visual receptive-field effects by using feedback predictions and feedforward residuals [5]. In behaving mice, visual-cortical populations combine locomotion and visual speed [11], and transient visuomotor mismatch responses are measurable during closed visuomotor behavior [10]. Random visuomotor coupling during development altered later Layer 2/3 responses, including inhibitory contributions [15]. Leinweber and colleagues identified a cortical route carrying locomotion-related visual-flow predictions [16]. These results make a central distinction concrete: a difference signal is meaningful relative to a learned expectation and a defined receiver, not because deviation is intrinsically informative in isolation.

The studies do not show that every cortical response is a prediction error, that the entire self is a predictive model, or that a mismatch is conscious. They establish component operations: expectations can be learned, motor-related variables can reach sensory cortex, and departures from expected sensorimotor relations can be measured.

3.2 Action return is typed, not a scalar reward

Corollary discharge and reafference research shows that the brain distinguishes sensory events caused by its own actions from externally caused events. Sommer and Wurtz identified a superior-colliculus-thalamus-frontal-eye-field pathway carrying information used across sequential saccades [9]. Schneider, Nelson, and Mooney found a motor-to-auditory cortical projection that recruits inhibitory circuitry before and during movement [12]. Later work showed experience-dependent suppression of predictable movement-related acoustic consequences [17]. In the cerebellum, Purkinje-cell populations combine motor-related and sensory information in a way that predicts the consequences of active self-motion [19].

Proprioceptive return is similarly distributed. Somatosensory area 2 contains neurons with different mixtures of active- and passive-movement sensitivity [18]. Spinal internal-copy and inhibitory circuits contribute to skilled movement and prevent destabilizing sensory feedback [13,14]. Visual and mechanical perturbations recruit partially overlapping primary-motor-cortex population subspaces during online correction [20], while interoceptive accuracy has been related to activity in insular and associated systems [8]. These findings justify typed return channels-visual, vestibular, proprioceptive, interoceptive, and motor-related prediction-rather than one undifferentiated “feedback” variable.

They also impose a control. A contingent return may differ from replay because of movement, arousal, reward, predictability, or temporal alignment. The experimental design must match those variables and preserve only the causal action-to-return relationship.

3.3 Phase can influence communication without being a universal carrier

Womelsdorf and colleagues found that the influence between neuronal groups depended on their phase relation, with effective relations specific in time, frequency, and space [21]. Bosman and colleagues reported attention-related selective synchronization between macaque visual areas [22]. Laminar recordings and interareal analyses found alpha- and gamma-associated directional tendencies in sampled visual circuits [23,24]. These results support a bounded communication-through-coherence claim: relative timing can modulate effective interaction among anatomically connected populations.

Phase can also carry information beyond spike count. Montemurro and colleagues found that phase-of-firing relative to low-frequency field activity conveyed information about natural visual stimuli not available from spike counts alone [25]. Theta-gamma coupling changed during item-context learning in rat hippocampus [26]. These results motivate receiver-relative timing features, but they do not prove SAN's PWD measure, identify a universal neural code, or imply that matching absolute frequency is sufficient for communication.

The paper therefore operationalizes “matching synaptic frequencies” as a family of testable receiver-relative relations: preferred phase offsets, conduction delays, synaptic and dendritic readiness, task-dependent frequency bands, and pathway-specific effective coupling. Two populations may communicate with a stable nonzero phase lag; two oscillations at the same nominal frequency may fail to communicate; and a broadband transient may matter without constituting a stationary oscillation.

3.4 Synchrony is not differentiated content or consciousness

Long-range gamma synchronization has correlated with perceptual organization and conscious report in human experiments [27,28]. Those results helped establish an important coordination program. Yet gamma is not uniquely sufficient for awareness. Pitts and colleagues found induced gamma and P3 effects that tracked post-perceptual task relevance rather than visual awareness under their contrasts [29].

SAN's proposed completion is therefore deliberately two-sided. Coherence can help make a route jointly available. A receiver-relative departure—an excitation-driven advance, inhibition-driven delay, missing event, phase shift, or structured splay—can differentiate what is occurring within that available route. Shannon's self-information provides the statistical lens: an event is informative relative to an expected distribution, not because desynchronization is inherently meaningful [4]. Biological consequence additionally requires a tuned receiver and a causal effect.

3.5 Learned networks can complete partial patterns

The recurrent connectivity of hippocampal CA3 motivated a pattern-completion test in which CA3 NMDA-receptor disruption selectively impaired retrieval from partial cues [30]. Optogenetic activation of neurons tagged during fear learning later evoked memory recall in mice [31]. These experiments show that partial-cue completion and causal reinstatement of learned ensembles are measurable.

They do not show that one engram is a complete self, that reinstatement reproduces a previous conscious experience exactly, or that the same mechanism explains all cortical rendering. In this paper, pattern completion is an experimental analogue for one edge: a partial cue can recruit more of a learned distributed state, and the causal route can be perturbed.

3.6 Closed interfaces reveal coadaptation and network constraint

Brain-computer interfaces make action-return mapping experimentally explicit. Long-term neuroprosthetic practice can stabilize a cortical control map [32]. Learning is easier for activity patterns within an existing neural manifold than for patterns outside it [33], while sufficiently extended learning can produce new activity patterns that causally support new behavior [35]. Decoder adaptation and user plasticity can shape one another [34].

These findings strongly motivate exact-versus-remapped rescue and the requirement to measure both sides of a coupled interface. They do not imply that any BCI becomes part of a conscious self or that an artificial controller is sentient. They establish that a fixed mapping, network history, and training duration constrain which new action-return relations are learned and retained.

3.7 Artificial world models are necessary comparators, not evidence of consciousness

Modern artificial agents can learn compact world models and act by predicting future outcomes. DreamerV3 learned across more than 150 control tasks with a common configuration and used a learned recurrent state-space model to support action selection [36]. This makes a world-model baseline mandatory for the proposed Artificial Neurology application. A full SAN-inspired

condition must beat or explain behavior beyond a strong recurrent world-model agent under matched capacity, data, optimization, action space, and compute.

Even a superior full condition would support only the measured implementation. World-model performance, self-report, recurrence, persistent state, or use of proprioceptive variables is not sufficient evidence of consciousness. The paper's strongest artificial claim is developmental: contingent action-return, mapping-specific receiver adaptation, and slow retained change jointly improve preregistered transfer or rescue endpoints beyond matched alternatives.

4. The proposed biological mechanism

4.1 The self is the loop, not one node

The self is not identified with one cortical area, frequency band, cell class, self-symbol, or fixed attractor. It is the continuing embodied route through which current sensory and body variables, learned dispositions, distributed activity, action, and returned consequences remain mutually consequential. A learned self-model can be active within this route, but it is not identical to the complete process.

4.2 Tonic context and phase wave differential

Let $z_i(t)$ denote a measured multivariate state at receiver i , including whichever phase, amplitude, rate, dendritic, synaptic, or population variables are declared before analysis. Let T_t be the expected tonic context and H_i the receiver's learned history and present readiness. Define the receiver-relative departure

$$\Delta z_i(t) = z_i(t) - \mathbb{E}[z_i(t) | T_t, H_i]. \quad (1)$$

Equation (1) is an operational family, not a claim that PWD is always one scalar phase difference. A valid measurement must declare the estimator, time scale, reference process, conditioning variables, and uncertainty. The corresponding conditional self-information for a discretized or otherwise properly density-normalized event $B_i(t)$ is

$$I_i(t) = -\log_2 P(B_i(t) | T_t, H_i). \quad (2)$$

Rarity is not meaning by itself. A high-surprisal fluctuation becomes biologically relevant only if a receiver is sensitive to it and an intervention on it changes a downstream result.

4.3 Phase-compatible route recruitment

For a directed sender-receiver edge $j \rightarrow i$, define a candidate compatibility score

$$C_{ij}(t) = w_{ij}(t)q_i(t) \cos(\phi_j(t - d_{ij}) - \phi_i(t) - \psi_{ij}), \quad (3)$$

where w_{ij} is effective connection strength, q_i is receiver readiness, d_{ij} is a conduction and processing delay, and ψ_{ij} is a preferred phase offset. Equal absolute phase is neither required nor sufficient. For a declared route P ,

$$R_P(t) = \sum_{(j,i) \in P} \rho_{ij}(t)[C_{ij}(t)]_+, \quad (4)$$

where ρ_{ij} records whether the edge is anatomically and functionally available. A causal claim requires that targeted manipulation of compatibility changes a preregistered downstream endpoint while rate, power, energy, movement, and other nuisance variables remain controlled.

4.4 Reinstatement without identity inflation

Let x_t be the current distributed state, m_k a learned reference pattern, and u_t a vector of nuisance features. A residualized reinstatement score can be written

$$S_k(t) = \text{sim}(x_t, m_k) - g(u_t), \quad (5)$$

where g is fitted only on the training partition. Partial-cue reinstatement is identified only if the unrepresented components of m_k are recovered above matched controls and an independently specified route intervention changes that recovery. Similarity does not establish experiential identity; it measures a relation between declared representations.

4.5 Fast state and slow learning

Let o_t be typed observation, b_t body state, x_t active state, a_t action, and θ_t slower receiver and route dispositions:

$$x_{t+1} = F_{\theta_t}(x_t, o_t, b_t, T_t), \quad (6)$$

$$a_t = \Pi_{\theta_t}(x_t), \quad (7)$$

$$(e_{t+1}, b_{t+1}, o_{t+1}) = G(e_t, b_t, a_t, \varepsilon_{t+1}), \quad (8)$$

$$\theta_{t+1} = U(\theta_t, x_t, a_t, o_{t+1}, b_{t+1}). \quad (9)$$

The system “learns itself” in the operational sense only when its prior state and action partly cause later input, returned consequences change θ , that change affects later recognition or action, and a named edge can be disrupted and specifically rescued. Activity-dependent homeostatic scaling supplies one established example of a slower change in neuronal response disposition, but it does not by itself establish this complete learning-loop operator [37].

5. The Gamma Consideration Sandwich as a bounded junction

The mature GCS mapping is:

1. **Layer 4 alpha, primary-sensory-side intake.** This is the proposed primary-sensory-side role.
2. **Layer 4 beta, prefrontal thought and top-down intake.** Alpha and beta are parallel intake-side roles; beta is not assigned the consideration operation.
3. **Layer 2/3 gamma, consideration and proprioceptive coordination.** This is the proposed functional middle.
4. **Layer 5 theta/gamma, action-side output and body-derived return.** This is the proposed action-return side.

The functional chain is:

```

Layer 4 alpha sensory-side input
+ Layer 4 beta prefrontal/top-down input
→ Layer 2/3 gamma consideration and proprioceptive coordination
→ Layer 5 theta/gamma action-side output and body-return state
→ changed body and world
→ proprioceptive, vestibular, interoceptive, and exteroceptive return
→ changed receiver readiness and learning

```

GCS is not established by the separate laminar and rhythm findings cited above. Those studies were performed in particular species, areas, layers, tasks, and frequency definitions. The complete sandwich must be tested as a source-receiver-direction-layer-body-action relation. Band power alone is insufficient.

A decisive GCS test must therefore measure or manipulate the four roles separately. It must compare the declared mapping with a role-scrambled mapping, control movement and arousal, preserve total stimulation energy, and distinguish exact restoration from an equally strong remapped rescue. If a simpler recurrent or motor-control model explains the endpoints, GCS has not earned added explanatory value in that implementation.

6. Artificial Neurology reference-application contract

The first application is a bounded virtual organism, not a simulation of human consciousness. The organism occupies a two-dimensional or three-dimensional environment with declared exteroceptive sensors, joint- or actuator-state channels, an energy/interoceptive channel, discrete or continuous actions, and a world whose response depends on those actions.

The full condition contains:

1. typed sensory and body-return channels;
2. a recurrent active self-world state;
3. receiver-specific transformations and a declared tonic reference;
4. a phase- or timing-compatible routing mechanism;
5. an action policy;
6. a slow, provenance-recorded update to receiver or route dispositions; and
7. exact interruption, remapping, retention, transfer, and rescue interfaces.

Capacity- and training-matched comparators must include at least:

- a feedforward policy;

- a recurrent world-model agent;
- a fixed-return controller;
- an open-loop replay condition;
- a yoked-return condition;
- a role-scrambled return condition;
- a no-phase/timing condition;
- a no-slow-learning condition; and
- a generic recurrent controller with the same parameter and compute budget.

The primary endpoints are mapping-specific transfer, prediction calibration, partial-cue completion, post-interruption recovery, exact-versus-remapped rescue, retained improvement after context change, and failure rate. Behavioral score alone cannot identify the proposed mechanism. Internal endpoints must test whether the declared active state, return channels, route variables, and slow updates actually mediated the result.

Development and held-out partitions must be generated and sealed before model comparison. Every negative, null, adverse, failed, or invalid result remains in the evidence package. The final claim class is selected from the evidence after the frozen analysis runs: implementation failure, mechanism not identified, bounded mechanism support, or externally replicated mechanism support. “Conscious” is not an available label in this application.

7. Hypotheses and causal experiments

H1. Closed-contingency advantage

Contingent action-return learning will change later prediction, transfer, or reinstatement more than a marginally and temporally matched yoked replay that is not caused by the system's current action.

H2. Timing-compatible recruitment

Receiver- and route-compatible timing will support more selective route use than timing-scrambled input matched for rate, power, amplitude, duration, and energy.

H3. Learned completion

A partial cue will recruit unrepresented components of a learned distributed pattern more strongly after relevant training, above matched familiarity, salience, and cue-strength controls.

H4. Mapping-specific loss and rescue

Breaking a named action-to-return or route relation will impair preregistered endpoints, and restoring the exact relation will rescue them more than restoring an equally strong remapped relation.

H5. Coupling is not content

Compatibility or coherence will better predict route availability, while receiver-relative departures and structured dispersion will better discriminate represented content. Global synchrony alone will not identify the self-world state.

H6. Learning changes future compatibility

Repeated contingent experience will alter receiver readiness, effective connectivity, preferred timing, or route use in a manner that predicts later behavior or reinstatement.

H7. Nested-loop dependence

Sensory, body-return, mnemonic, and action loops will show specific interaction effects. Removing one loop will produce a predicted deficit rather than only global performance degradation.

H8. GCS role-specific closure

After controlling for movement, arousal, rate, power, anatomy, and history, the declared sensory-side, context-side, consideration, and action-return variables will jointly predict correction or updating. Exact mapping rescue will outperform role-scrambled rescue.

H9. Artificial developmental retention

The full artificial condition will retain mapping-specific improvement across a declared context change more strongly than capacity-matched fixed, open-loop, yoked, scrambled, and no-slow-learning controls.

8. Identifiability preview

The mathematical and application drafts must distinguish four questions:

1. **Observable equivalence:** do two systems produce indistinguishable measured outputs within a declared tolerance?
2. **Causal identification:** does intervening on the proposed edge change the endpoint relative to appropriate controls?
3. **Implementation adequacy:** does the code instantiate the variables and transitions claimed in the model?
4. **Consciousness interpretation:** does the evidence warrant a claim about experience, identity, or welfare?

The first three can be addressed in bounded synthetic and biological experiments. The fourth cannot be inferred from them automatically.

For contingent return, let Y be a preregistered retention or transfer endpoint. The primary estimand is

$$\tau_{\text{closed}} = \mathbb{E}[Y \mid \text{do}(C = 1)] - \mathbb{E}[Y \mid \text{do}(C = 0)], \quad (10)$$

where $C = 0$ is a yoked condition matched on the frozen nuisance contract. For exact versus remapped rescue,

$$\tau_{\text{map}} = \mathbb{E}[Y \mid \text{do}(M = M_{\text{exact}})] - \mathbb{E}[Y \mid \text{do}(M = M_{\text{remap}})]. \quad (11)$$

For the timing contribution, a rate- and power-controlled contrast is

$$\tau_{\text{time}} = \mathbb{E}[Y \mid \text{do}(Q = Q_{\text{compatible}})] - \mathbb{E}[Y \mid \text{do}(Q = Q_{\text{scrambled}})]. \quad (12)$$

None of these estimands identifies the whole theory by itself. A full result requires a preregistered conjunction across behavior, internal mediation, mapping specificity, transfer, and failure boundaries. The following mathematical section supplies the causal graph, countermodels, uncertainty, multiplicity, dependence, power, and nonidentifiability arguments.

9. Mathematical and causal identification

9.1. Four inferential layers

The paper separates four conclusions that are often conflated.

1. **Observable equivalence.** Two systems are equivalent on a declared observation map and tolerance. Hidden implementation and experience may still differ.
2. **Causal identification.** A defined intervention on one edge changes a preregistered endpoint relative to appropriate controls.
3. **Implementation adequacy.** The executable system instantiates the variables, permissions, transitions, and provenance contract claimed for it.
4. **Consciousness interpretation.** Evidence warrants a claim about experience, welfare, or identity.

the causal, implementation, and development studies can test the first three in a bounded synthetic environment. They do not automatically license the fourth.

9.2. Typed structural causal model

Let experimental unit i be a seed-defined environment-agent pairing and t a within-unit tick. The baseline variables are environment family E_i , target/context K_i , initial body state B_{i0} , initial fast state X_{i0} , initial slow disposition Θ_{i0} , and randomized condition C_i . Within a tick:

- O_{it}^s : primary-sensory-side observation;
- O_{it}^c : top-down/context observation;
- B_{it} : typed proprioceptive/body-return state;
- X_{it} : fast active/consideration state;
- Q_{it} : receiver-relative phase-differential feature;
- L_{it} : route-compatibility and readiness variables;

- A_{it} : action;
- $R_{i,t+1}$: returned consequence;
- Θ_{it} : slower receiver, route, threshold, and memory dispositions; and
- Y_i : episode- and transfer-level outcomes.

The causal factorization is

$$\begin{aligned}
p(V_i) = & p(E_i, K_i, B_{i0}, X_{i0}, \Theta_{i0}) p(C_i) \prod_{t=0}^{T-1} p(O_{it}^s, O_{it}^c \mid E_i, K_i, B_{it}) p(Q_{it}, L_{it} \mid O_{it}^s, O_{it}^c, B_{it}, X_{it}, \Theta_{it}, C_i) \\
& \times p(X_{i,t+1} \mid X_{it}, O_{it}^s, O_{it}^c, B_{it}, Q_{it}, L_{it}, \Theta_{it}, C_i) p(A_{it} \mid X_{i,t+1}, \Theta_{it}, C_i) \\
& \times p(E_{i,t+1}, B_{i,t+1}, R_{i,t+1} \mid E_{it}, B_{it}, A_{it}, C_i) p(\Theta_{i,t+1} \mid \Theta_{it}, X_{i,t+1}, A_{it}, R_{i,t+1}, C_i) p(Y_i \mid V_{i,0:T}).
\end{aligned} \tag{13}$$

Randomization occurs at the seed-defined unit, not at the event row. Event rows are repeated measures and are never treated as independent experimental units.

9.3. Fast state, slow disposition, and body/world closure

The fast update is a bounded recurrent operator

$$X_{t+1} = (1 - \lambda_x) X_t + \lambda_x \tanh(W_s O_t^s + W_c O_t^c + W_b B_t + W_q Q_t + W_r R_t + W_x X_t), \quad 0 < \lambda_x \leq 1. \tag{14}$$

The action policy is

$$A_t = \arg \max_{a \in \mathcal{A}} \{u_a^\top X_{t+1} + v_a^\top O_t^c - \kappa \text{cost}(a, B_t)\}, \tag{15}$$

with a deterministic tie rule frozen in the implementation contract. Environment and body transition jointly:

$$(E_{t+1}, B_{t+1}, R_{t+1}) = G(E_t, B_t, A_t, \epsilon_{t+1}). \tag{16}$$

The slow update is

$$\Theta_{t+1} = \Pi_\Omega \left[\Theta_t + \eta m_t \nabla_{\Theta} \ell \left(\hat{R}_{t+1}(X_t, A_t; \Theta_t), R_{t+1} \right) \right], \quad 0 < \eta \ll \lambda_x, \tag{17}$$

where m_t is a permission and provenance mask and Π projects onto a declared compact parameter set Ω . Every update records parent state, exposure identity, old value, proposed value, accepted value, and rollback token. A fast activation change is not counted as learning; a slow change without exposure provenance is invalid.

9.3.1. Boundedness

Proposition 1 (fast-state boundedness). If $X_0 \in [-1, 1]^d$, $0 < \lambda_x \leq 1$, and the nonlinear candidate in Equation 14 uses componentwise $\tanh$, then $X_t \in [-1, 1]^d$ for all finite t .

Proof. The candidate is in $[-1, 1]^d$. Equation 14 is a convex combination of X_t and that candidate. Induction preserves the box. $\square$

Proposition 2 (slow-disposition boundedness). If Ω is closed and bounded and every slow update is projected by Π_Ω , then $\Theta_t \in \Omega$ for every accepted update.

Proof. Projection maps every proposed value into Ω . Rejected edits retain the prior value, already in Ω . $\square$

9.3.2. Timescale criterion

The implementation claims two operational timescales only if all three conditions hold:

$$\rho_x = \text{median}(\|X_{t+1} - X_t\|_2) > 0, \quad \rho_\Theta = \text{median}(\|\Theta_{t+1} - \Theta_t\|_2) > 0, \quad \frac{\rho_\Theta}{\rho_x} \leq \delta_{\text{slow}}, \tag{18}$$

and Θ remains changed after fast state reset. The reference implementation freezes $\delta_{\text{slow}} = 0.20$ for the bounded reference implementation. This is an engineering separation, not a claim that biological timescales have the same ratio.

9.4. Receiver-relative PWD and route identifiability

Let ϕ_{jt} be the phase assigned to event component (j), $\mu_j(r, h)$ its training-frozen expected phase under receiver (r) and context-history stratum (h), and $\sigma_j(r, h) > 0$ a frozen scale. Define the signed wrapped residual

$$q_{jrt} = \frac{\text{wrap}_{(-\pi, \pi)}(\phi_{jt} - \mu_j(r, h_t))}{\sigma_j(r, h_t)}. \tag{19}$$

PWD is the vector $Q_{rt} = (q_{1rt}, \dots, q_{prt})$, not a scalar synonym for variance, desynchronization, surprise, or information. A training-frozen finite partition $\mathcal{P}$ maps continuous PWD values into bins for conditional self-information:

$$I_{rt} = -\log_2 \hat{P}_{\text{train}}(\mathcal{P}(Q_{rt}) \mid r, h_t). \quad (20)$$

No rescue epsilon is inserted into an unseen bin. Unseen bins receive an explicit `OUT_OF_SUPPORT` classification and are excluded from claims requiring calibrated surprisal.

Route (k) receives the score

$$S_{krt} = \alpha_k - \beta_k d_{\text{phase}}(Q_{rt}, \nu_{kr}) + \gamma_k^\top X_t + \zeta_k^\top B_t - \omega_k H_{krt}, \quad (21)$$

where ν_{kr} is the learned receiver-route template and (H) is inhibition or unavailability. Route use is a training-frozen softmax or exact argmax specified per condition.

9.4.1. Route-identification criterion

Timing contributes an identified mechanism only if:

1. rate, event count, marginal power, anatomy, movement, reward, context, and exposure are matched;
2. a phase-compatible intervention changes preregistered behavior or reinstatement relative to a phase-scrambled intervention;
3. an exact receiver-route remapping rescues more than an arbitrary or wrong remapping;
4. the effect remains after a rate-only and residual-only comparator; and
5. the internal route variable mediates the effect under a preregistered mediation analysis without being adjusted away in the total-effect model.

9.5. Potential outcomes and estimands

Let $(Y_i(c))$ be the episode endpoint under condition (c). The primary total effects are paired over seed-defined worlds.

Closed contingency versus yoked replay:

$$\tau_{CY} = \mathbb{E}[Y_i(C_{\text{FULL}}) - Y_i(C_{\text{YOKED}})]. \quad (22)$$

Closed contingency versus open loop:

$$\tau_{CO} = \mathbb{E}[Y_i(C_{\text{FULL}}) - Y_i(C_{\text{OPEN}})]. \quad (23)$$

Exact versus wrong-map rescue after route disruption:

$$\tau_{MR} = \mathbb{E}[Y_i(C_{\text{EXACT_RESCUE}}) - Y_i(C_{\text{WRONG_RESCUE}})]. \quad (24)$$

Timing-compatible versus phase-scrambled return:

$$\tau_{QS} = \mathbb{E}[Y_i(C_{\text{FULL}}) - Y_i(C_{\text{PHASE_SCRAMBLED}})]. \quad (25)$$

Slow learning versus frozen dispositions:

$$\tau_{\Theta} = \mathbb{E}[Y_i(C_{\text{FULL}}) - Y_i(C_{\text{NO_SLOW}})]. \quad (26)$$

PWD versus rate/residual-only matched representation:

$$\tau_{PWD} = \mathbb{E}[Y_i(C_{\text{FULL}}) - Y_i(C_{\text{NO_PWD}})]. \quad (27)$$

The bounded full-mechanism gate requires the directionally favorable conjunction of Equations 22-27 on their registered endpoints, not merely one small p-value.

9.6. Post-treatment variables and mediation

Fast state, route use, PWD-derived scores, update count, energy use, and movement can be consequences of condition. They must not be inserted automatically as baseline covariates in the total-effect model. The total effect uses randomized condition and baseline strata only. A separate mediation model may estimate the interventional indirect effect through route use (M):

$$\tau_{\text{route}} = \mathbb{E}[Y(C_{\text{FULL}}, M(C_{\text{FULL}})) - Y(C_{\text{FULL}}, M(C_{\text{SCRAMBLED}}))], \quad (28)$$

subject to declared sequential-exchangeability assumptions that are exactly satisfied only in the simulator. The biological analogue would require stronger design and cannot inherit identification from the synthetic model.

9.7. Nonidentifiability results and countermodels

Countermodel 1: self-reference without self-learning

Two agents emit identical self-descriptions. Agent A contains a fixed string “I am this agent”; Agent B has an action-return learner. Under a self-report-only observation map they are equivalent although only B updates from self-caused return. Therefore self-reference does not identify self-learning.

Countermodel 2: self-model without whole-loop closure

An open-loop predictor receives a body-state vector and accurately estimates its location. A closed-loop agent does the same while its actions determine later input. Static body-model accuracy can be identical. Therefore self-model accuracy does not identify action-return closure.

Countermodel 3: common-input synchrony

Let $Q_t = U_t$ and $Y_t = U_t + \epsilon_t$ for an unobserved common driver U_t , with no edge $Q \rightarrow Y$. Correlation and phase locking may be arbitrarily high while the causal effect of intervening on Q is zero. Therefore synchrony and predictive association do not identify phase-mediated influence.

Countermodel 4: firing-rate confound

A phase manipulation also changes spike/event count, and the endpoint depends only on count. Phase-compatible and scrambled groups differ even though phase has no independent effect. Rate and marginal-power matching are necessary.

Countermodel 5: replay equivalence

If a policy is fixed and the replayed return sequence equals the contingent sequence, contingent and yoked trials can be observationally identical. Identification requires divergence-producing interventions or later adaptation, not passive replay alone.

Countermodel 6: arbitrary rescue

If any added route increases capacity, exact-map and wrong-map rescue can both improve. A rescue claim requires exact mapping to exceed capacity-matched wrong and random maps.

Countermodel 7: label permutation

Two internal route labels can be permuted with compensating decoder changes and identical behavior. Route identity is identifiable only up to a declared equivalence class unless interventions anchor it to inputs and consequences.

Countermodel 8: timescale aliasing

A rapidly decaying fast state can appear retained when measurements are too close together; a slow update can appear absent when evaluation occurs after overwriting. Fast reset and delayed transfer are required.

Countermodel 9: tonic-distribution leakage

If the expected phase distribution is fitted using evaluation events, surprisal and PWD thresholds leak outcome structure. The tonic model and partition must be trained and frozen without evaluation access.

Countermodel 10: architecture privilege

If the full model receives body-return or context variables unavailable to the comparator, an apparent mechanism advantage is an information-set advantage. All admissible systems receive identical observable inputs; only the registered operation differs.

Countermodel 11: capacity privilege

If the full model has more tunable state or compute, it may win generically. Parameter/state/operation counts and wall-resource proxies must be reported, and both capacity-matched compact controls and stronger architecture-native controls must be retained.

Countermodel 12: successful behavior without the declared internal mechanism

A memorizing lookup controller can solve a finite environment while its purported PWD and route variables are unused. Intervention and mediation gates must verify causal participation, not mere presence in code.

9.8. Statistical unit, uncertainty, and multiplicity

The seed-defined world-agent pairing is the primary independent unit. Ticks, event rows, routes, and episodes nested under one seed are repeated measures.

For contrast j , paired seed difference D_{ij} gives

$$\hat{\tau}_j = \frac{1}{n} \sum_{i=1}^n D_{ij}, \quad \widehat{SE}_{\text{cluster}}(\hat{\tau}_j) = \sqrt{\frac{1}{n(n-1)} \sum_{i=1}^n (D_{ij} - \hat{\tau}_j)^2}. \quad (29)$$

The primary interval is a 95% percentile bootstrap over complete seed clusters with 5,000 frozen resamples. A paired randomization/sign-flip test over seed differences supplies the primary p-value when exchangeability holds. Exact enumeration is used when $n \leq 20$; otherwise 100,000 frozen sign draws are used.

The six primary mechanism contrasts in Equations 22-27 form one Holm family. A result is a **complete bounded support** only when all six adjusted tests reject in the registered direction and every practical-effect floor passes. Internal mediation and secondary task endpoints form a separate Benjamini-Hochberg family at ($q=0.05$) and cannot rescue a failed primary gate. Safety, invalidity, leakage, replay, and provenance checks are conjunctive fail-closed gates, not multiplicity-adjusted discoveries.

9.9. Power and precision freeze

Development Study 1 is a development pilot, not the held-out final study. It uses eight development seeds to expose implementation defects and estimate variance without making a confirmatory claim. A later held-out synthetic evaluation targets 24 complete paired seeds per condition. Before access, the program computes the minimum detectable standardized paired effect under a two-sided familywise α allocation and reports the exact value. No observed pilot mean may be used to relax an effect floor.

The practical floors are frozen in endpoint-native units:

- success/transfer probability: +0.05;
- calibrated return-prediction score: +0.03;
- partial-cue reinstatement: +0.05;
- exact-minus-wrong rescue: +0.05;
- post-interruption recovery: +0.05;
- retained mapping-specific improvement: +0.05.

If the 95% interval includes the floor or the adjusted p-value fails, the contrast is unresolved even if the point estimate is favorable. If the interval lies below zero, it is adverse. Null, adverse, and infrastructure-failure rows remain in all later drafts.

9.10. Frozen reference-application conditions

The reference application must implement these conditions with the same environment seeds, observation set, action set, episode budget, and result schema:

Code	Condition	Distinguishing operation
C0	FULL_LOOP	Contingent typed return, PWD, route compatibility, fast state, and slow updates
C1	FIXED_CONTROLLER	Same interface; frozen policy and no retained learning
C2	OPEN_LOOP	Return cannot affect later state or disposition
C3	YOKED_REPLAY	Marginal return sequence replayed from another seed; no action contingency
C4	PHASE_SCRAMBLLED	Same events/rate/power; receiver-relative timing assignment scrambled
C5	NO_PWD	Rate/residual-only representation with matched state capacity
C6	NO_SLOW	Fast state active; Θ frozen
C7	WRONG_MAP_RESCUE	Disrupted route receives capacity-matched incorrect remapping
C8	EXACT_MAP_RESCUE	Disrupted route receives the exact registered remapping
C9	GENERIC_RECURRENT	Same observables and declared scalar-state capacity; generic recurrent update
C10	LOOKUP_ORACLE	Environment-specific upper-bound comparator; not admissible as a learning model

The lookup oracle establishes task ceiling and leakage detection. It is never counted as evidence for or against the proposed learning operation.

9.11. Application task family

The reference environment is a bounded embodied grid with hidden context-dependent return mappings. Each seed defines:

- a 9×9 world;
- one target and two decoys;
- local primary-sensory cues;
- a top-down target-context cue;
- body heading, movement success, collision, and energy return;
- phase-tagged event components generated from a training-frozen tonic template;
- a context switch;
- a route disruption;
- exact and wrong-map rescue fixtures;
- partial-cue probes; and
- delayed transfer after fast-state reset.

The system must learn which receiver-route mapping makes sensory, context, and body return jointly predictive. No natural-language self-report is used as a primary endpoint.

9.12. Event and provenance contract

Every event must contain:

1. schema version;
2. run, seed, condition, episode, and tick identifiers;
3. environment and context identifiers;
4. sensory, context, body-return, PWD, route, fast-state, action, and consequence fields;
5. parent-state hash and resulting-state hash;
6. slow-update proposal, permission, acceptance, and rollback fields;
7. model/config/source hashes;
8. monotonic sequence number and timestamp-free deterministic ordering; and
9. validity and refusal codes.

No event may contain a result computed from future ticks. Replay must reproduce canonical scientific outputs byte-for-byte; resource timing may be recorded separately because wall time is not deterministic.

9.13. Gate logic

Let (G_m) be the six primary mechanism floors, (G_v) schema/provenance validity, (G_r) deterministic replay, (G_l) leakage absence, (G_c) capacity/resource disclosure, and (G_s) stability/boundedness. The complete bounded gate is

$$G_{\text{complete}} = \left(\bigwedge_{m=1}^6 G_m \right) \wedge G_v \wedge G_r \wedge G_l \wedge G_c \wedge G_s. \quad (30)$$

Passing G_{complete} would support the declared synthetic mechanism under the tested generator. It would not prove the full biological SAN theory. Failure identifies the exact unmet conjunction and remains publishable evidence.

10. Typed executable reference application

Learning-Loop Reference Application: System Context

Bounded synthetic mechanism test — not a biological or consciousness model

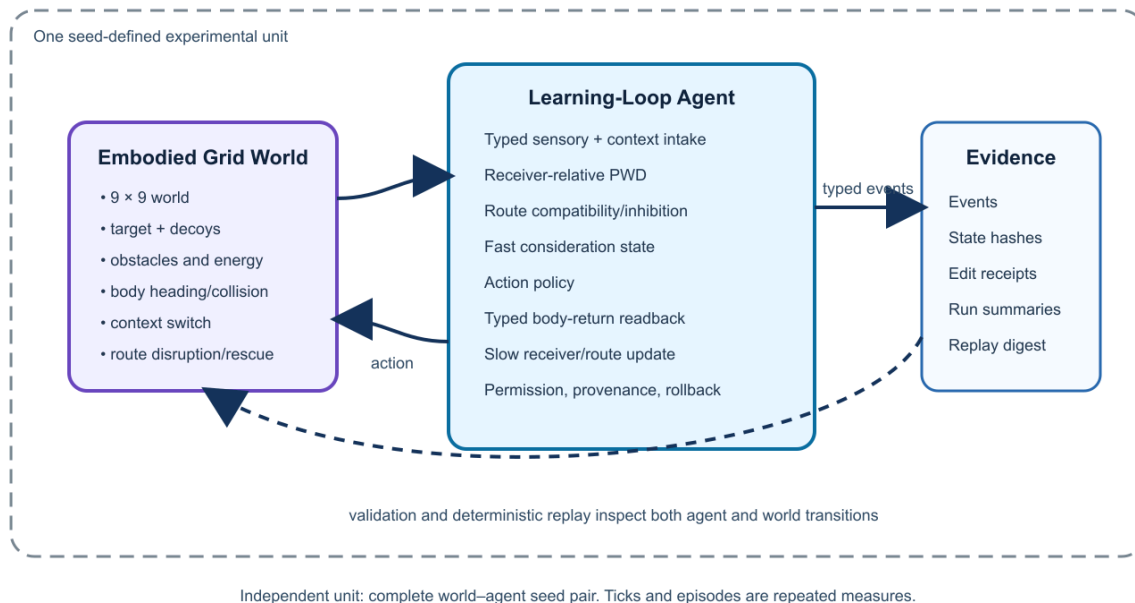


Figure 1. System context for the typed finite reference application.

Learning-Loop Component Architecture

Functional GCS-inspired roles are typed interfaces, not claimed biological layer/frequency validation

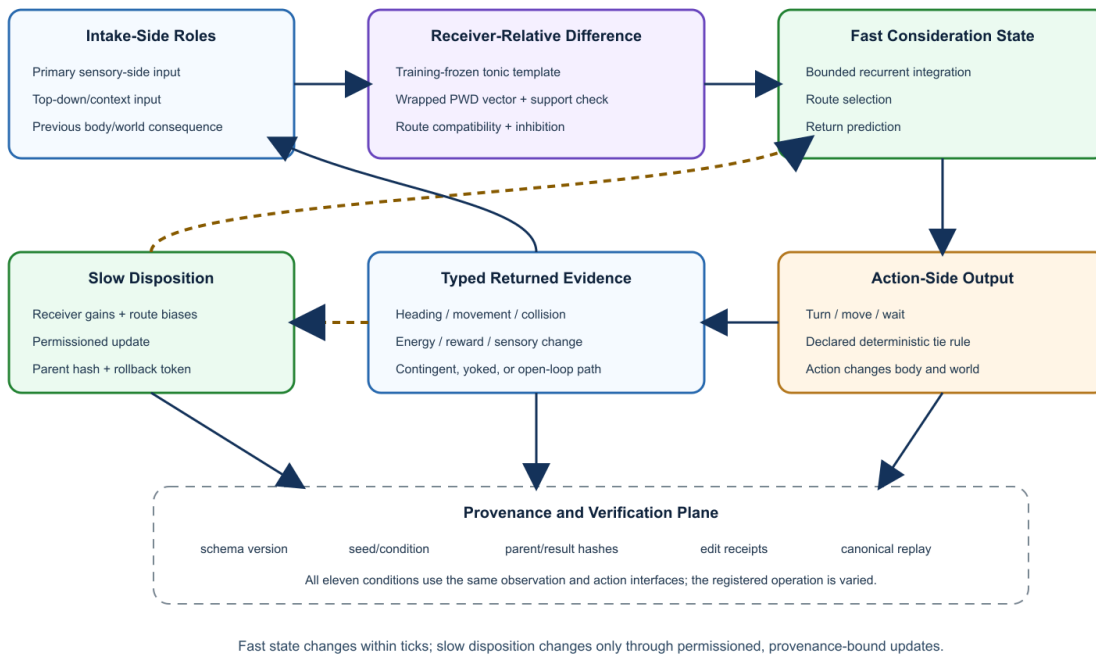


Figure 2. Functional component architecture. The roles are engineering analogues, not a cortical simulation.

10.1. System context

The experimental unit couples one seeded 9×9 world to one condition-specific agent. The world contains a target, two decoys, obstacles, heading, energy, collision state, a top-down context, phase-tagged events, a context switch, and disruption/rescue fixtures. The agent receives four typed input groups:

1. primary-sensory-side input;
2. top-down/context input;
3. proprioceptive/body return; and
4. the previous body/world consequence.

It exposes six typed internal interfaces:

1. receiver-relative PWD;
2. route compatibility;
3. route inhibition/unavailability;
4. a bounded fast consideration state;
5. a slower receiver/route disposition; and
6. a memory/update trace.

The system-context diagram is shown in Figure 1. The component diagram is shown in Figure 2.

10.2. Functional GCS-inspired mapping

The application instantiates functional roles rather than biological layers or rhythms. Primary sensory and top-down context are parallel intake-side interfaces. The receiver-relative PWD, route score, and bounded recurrent state form a consideration interface. Action changes the embodied world. Heading, movement, collision, energy, and consequence return through typed channels. A permissioned slower update changes route bias and receiver gain.

This organization tests whether the causal operations can be implemented and perturbed. It does not validate the biological assertion that Layer 4 alpha and beta, Layer 2/3 gamma, and Layer 5 theta/gamma realize those roles.

10.3. Common condition interface

All admissible conditions receive the same observation and action types. The registered operation varies:

- `FULL_LOOP` uses contingent return, PWD, route compatibility, fast recurrence, and slow change.
- `FIXED_CONTROLLER` uses the interface but denies slow updates.
- `OPEN_LOOP` prevents returned consequence from changing retained state.
- `YOKED_REPLAY` substitutes deterministic noncontingent body-return values.
- `PHASE_SCRAMBLLED` preserves the PWD marginal set while changing receiver assignment.
- `NO_PWD` zeros PWD while retaining matched state shape.
- `NO_SLOW` retains fast state and denies slow updates.
- `WRONG_MAP_RESCUE` supplies an incorrect route remapping after disruption.
- `EXACT_MAP_RESCUE` restores the registered route mapping.
- `GENERIC_RECURRENT` uses the same observables and an unstructured recurrent score.
- `LOOKUP_ORACLE` directly uses target geometry as a ceiling/leakage control and is not a learning model.

The executable condition operator is

$$\mathcal{A}_c : (O_t^s, O_t^c, B_t, R_t, X_t, \Theta_t) \mapsto (Q_t, L_t, X_{t+1}, A_t, \Theta_{t+1}), \quad c \in \mathcal{C}_{11}. \quad (31)$$

Interface equality means equal field availability and shape, not identical internal computation or exact parameter count. Later performance drafts must disclose state and operation counts and retain stronger controls.

10.4. Provenance-complete transaction

Before a tick, the application hashes the world and agent parent state. It then computes observation, PWD, route scores, fast state, action, consequence, and a proposed slow update. The slow update is accepted only if the condition permits it and the event carries nonempty exposure provenance. Each event records the parent hash, proposed edit hash, resulting slow-state hash, rollback token, and full result-state hash.

An event is accepted only if

$$V(e) = V_{\text{schema}} \wedge V_{\text{shape}} \wedge V_{\text{finite}} \wedge V_{\text{bound}} \wedge V_{\text{hash}} \wedge V_{\text{route}}. \quad (32)$$

No event contains a future outcome or next-event field. Deterministic sequence numbers replace wall-clock timestamps in the scientific projection. Wall-time resource measurements are kept outside the canonical digest.

10.5. Exact replay

The fixture runner executed 33 seed-condition projections: three fixture seeds across eleven conditions. The result contained 963 validated event rows. A clean replay reproduced all three canonical files byte-for-byte:

$$H(F_{\text{primary}}) = H(F_{\text{replay}}) \quad \text{for each of 3 canonical scientific files.} \quad (33)$$

The primary scientific-projection SHA-256 is `A02B29F184A9E544C7016ACFBD77F95BF9746C0986B586C2711606709AC72D2D`. The compact receipt SHA-256 is `EB69DB4355DF542B3B3511EDDB88A6E459EE0CC3D46D690A1C0841F9EF6D7D40`.

10.6. Tests

The 39 passing tests cover:

- deterministic world generation and replay;
- event shapes, finite values, bounds, hashes, and sequence order;
- absence of future-result fields;
- all eleven conditions;
- PWD zeroing and phase scrambling;
- permissioned and denied slow edits;
- exact rollback;
- slow persistence across episode calls;
- exact-versus-wrong rescue distinction;
- the oracle interface; and
- all twelve the frozen causal nonidentifiability countermodels.

The countermodel tests are executable witnesses that a weaker observation can be compatible with two different mechanisms. They are not empirical neuroscience results.

10.7. Retained implementation-fixture outcomes

The fixture outcomes are retained precisely because they are not uniformly favorable.

Condition	Successes / 3	Event rows	Mean fixture Brier loss
Full loop	0 / 3	96	0.6754
Fixed controller	0 / 3	96	0.6637
Open loop	0 / 3	96	0.6757
Yoked replay	0 / 3	96	0.6754
Phase scrambled	0 / 3	96	0.6750
No PWD	0 / 3	96	0.6832
No slow learning	0 / 3	96	0.6757
Wrong-map rescue	0 / 3	96	0.6944
Exact-map rescue	3 / 3	51	0.5380
Generic recurrent	0 / 3	96	0.4865
Lookup oracle	2 / 3	48	0.5179

These values do not pass or fail the Development Study 1 primary effects. The fixture seeds were selected for code-path coverage, use one disrupted episode per condition, and do not exercise the full training, reset, transfer, and uncertainty plan. The 0/3 full-loop result prevents an implementation pass from being rhetorically converted into performance support.

10.8. Known limitations before the pilot

1. The heuristic full route score is not yet calibrated for navigation performance.
2. Fixture runs use a single episode; retained learning is tested mechanically across calls but not evaluated scientifically here.
3. Scalar-state counts are bounded but not yet matched exactly across every condition.
4. The generic recurrent condition is a deterministic compact control, not a trained modern sequence model.
5. The lookup oracle can fail within the tick limit because it has direct target direction but still obeys obstacles and the action set; it is not a perfect planner.
6. Phase tags are synthetic and generated from the declared tonic template.
7. The application does not implement real cortical anatomy, synapses, biological oscillations, interoception, experience, or welfare.

11. Development Study 1: shortcut discovery and adverse evidence

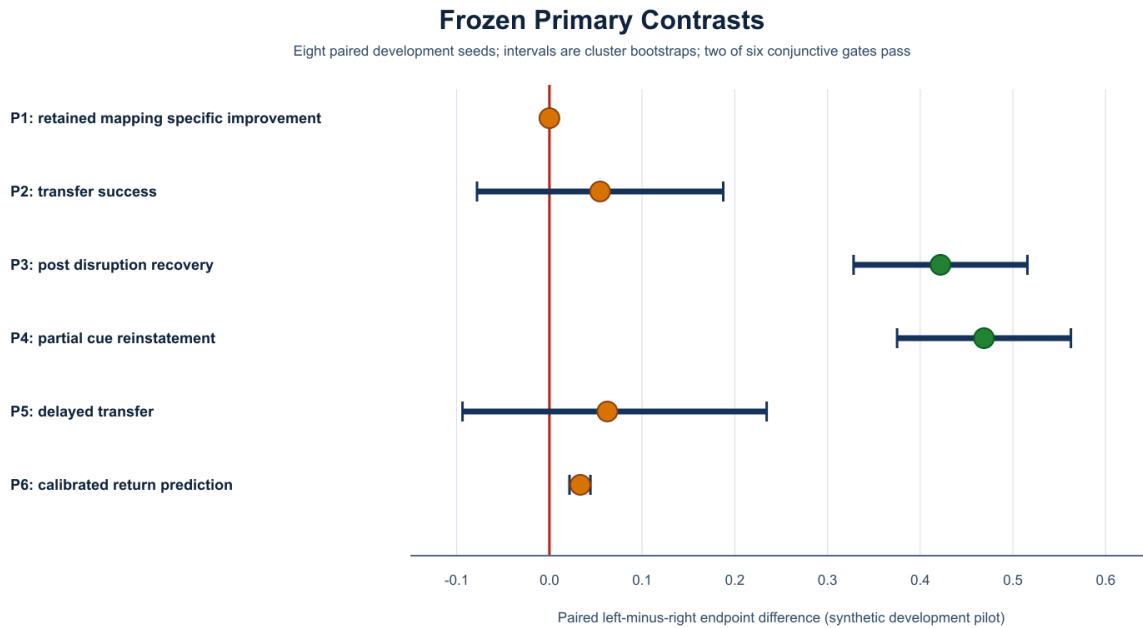


Figure 3. Frozen paired contrasts from Development Study 1; only two of six gates pass.

Condition-Level Development Outcomes

Means over eight seeds; display retains favorable, null, and adverse conditions

	Evaluation	Transfer	Partial cue	Recovery	Brier skill
FULL LOOP	0.495	0.414	0.516	0.000	0.409
FIXED CONTROLLER	0.042	0.023	0.047	0.000	0.337
OPEN LOOP	0.292	0.359	0.297	0.000	0.411
YOKED REPLAY	0.495	0.414	0.516	0.000	0.409
PHASE SCRAMBLED	0.000	0.000	0.047	0.000	0.325
NO PWD	0.089	0.078	0.062	0.000	0.376
NO SLOW	0.292	0.359	0.297	0.000	0.411
WRONG MAP RESCUE	0.495	0.414	0.516	0.000	0.409
EXACT MAP RESCUE	0.495	0.414	0.516	0.422	0.409
GENERIC RECURRENT	0.010	0.000	0.047	0.000	0.419
LOOKUP ORACLE	0.495	0.414	0.516	0.422	0.391

Cell color is descriptive only; inferential decisions use the frozen paired contrasts, not this heatmap.

Figure 4. Descriptive condition means from Development Study 1.

11.1. Design

Each of eight development seeds was run under all eleven frozen conditions. One agent persisted through 104 episodes:

- 48 training episodes;
- 24 evaluation episodes;
- 16 transfer episodes after an explicit fast-state reset;
- eight partial-cue episodes with directional sensory components removed; and
- eight route-disruption/rescue episodes.

The agent's slow disposition persisted across episodes. Fast state was reset before transfer. Conditions, endpoints, effect floors, independent units, bootstrap count, paired sign-flip tests, and Holm correction were frozen in the frozen causal specification before these outcomes existed.

11.2. Data integrity and replay

All 88 seed-condition runs completed. The 9,152 episode rows aggregate 456,330 schema-valid event transitions. No NaN, infinity, invalid shape, unknown route, missing hash, or runtime failure was recorded. Primary and replay files match byte-for-byte:

Artifact	Bytes	SHA-256
Run summaries	29,010	517A4024AEA9B4CA8E64C78085D8C50B278E4902CC494121AAAA32BA644844B4
Episode results	2,078,208	B4EFB826502D848DD42C394C2909DFD6D21E6D5579DB43E6492625103DC7D110
Primary contrasts	2,375	62B2D9BA9CE1C950CBFA381245A15120F2793BB969AFF0DFE204E0E869153AAE
Compact receipt	720	19AB6810314F935AD14747C67A155E5940DF650342260EA888E4B71767190D86

The canonical run, episode, and contrast digests are respectively

9c708d0cccff203431e8e498d7016db4decc389de290c8a821df881ea23091dc,
2c056b2ffa3936fffd6e9c76e6873600c3c00bc2c82b1638da3ea82c31aec4c5a, and

11.3. Primary results

The paired seed-level estimator is

$$\hat{\tau}_j = \frac{1}{8} \sum_{i=1}^8 (Y_{ij}^{left} - Y_{ij}^{right}). \quad (34)$$

Intervals are 5,000-resample percentile bootstraps over complete seed pairs. Unadjusted p-values are exact two-sided paired sign-flip tests; the six p-values form one Holm family. An endpoint passes only if its confidence interval clears the frozen practical floor and its Holm-adjusted p-value is below 0.05.

ID	Registered contrast	Mean difference	95% interval	Holm p	Verdict
P1	Full - yoked, retained improvement	0.0000	[0.0000, 0.0000]	1.0000	Exact null
P2	Full - open, transfer success	0.0547	[-0.0781, 0.1875]	1.0000	Unresolved
P3	Exact - wrong rescue, recovery	0.4219	[0.3281, 0.5156]	0.0469	Pass
P4	Full - phase scrambled, partial-cue success	0.4688	[0.3750, 0.5625]	0.0469	Pass
P5	Full - no slow, delayed transfer	0.0625	[-0.0938, 0.2344]	1.0000	Unresolved
P6	Full - no PWD, calibrated return prediction	0.0333	[0.0217, 0.0442]	0.0625	Conjunction fail

P6's point estimate exceeds the 0.03 floor, but the lower interval does not and the Holm-adjusted p-value is 0.0625. It is not a pass.

The complete gate is

$$G_{complete} = 0, \quad \sum_{j=1}^6 \mathbf{1}(G_j = 1) = 2. \quad (35)$$

This arithmetic is reported as “two of six gates passed,” not “two significant successes prove the theory.”

11.4. Condition-level description

The full loop's mean evaluation success was 0.4948, transfer success 0.4141, partial-cue success 0.5156, recovery after disruption 0, and evaluation Brier skill 0.4093. Exact rescue recovered 0.4219 of disruption episodes; wrong rescue recovered none. Phase scrambling reduced evaluation and transfer success to zero and partial-cue success to 0.0469. No-PWD evaluation success was 0.0885. These large descriptive differences motivate the two passing registered contrasts.

However, yoked replay was numerically identical to the full loop on evaluation, transfer, partial-cue, and Brier-skill means. Open loop and no-slow learning were also identical to each other on those means. The generic recurrent comparator obtained evaluation success 0.0104 and transfer success 0 but slightly higher Brier skill, 0.4189, than the full loop's 0.4093. The lookup oracle matched the full loop's non-disruption success profile and exact rescue's recovery, showing that the synthetic PWD route largely exposes the geometry needed by the heuristic action policy.

11.5. What the two passing contrasts identify

11.5.1. Mapping-specific rescue

Exact remapping versus wrong remapping changed post-disruption recovery by 0.4219. Both conditions used the same interface and state shapes. The result identifies mapping specificity in this implementation: arbitrary added route capacity did not reproduce the exact route mapping.

It does not establish biological receiver tuning, synaptic remapping, body incorporation, or conscious continuity.

11.5.2. Receiver-relative phase assignment

The full condition exceeded the phase-scrambled condition on the partial-cue endpoint by 0.4688. The manipulation permuted the same PWD values across receiver routes, preserving their marginal set within an event while changing compatibility. This identifies use of the synthetic phase-route assignment in the implemented policy.

It does not show that biological phase relations instantiate the same code, that gamma is conscious content, or that PWD is a universal neural operator.

11.6. What the failed contrasts reveal

11.6.1. Closed versus yoked: exact null

P1 is the most important negative result. The full and yoked conditions were exactly equal on the registered retained-improvement endpoint. The route score is dominated by phase-tagged target-direction information and slow updates driven by consequences available in both paths. Replacing focal body return with a deterministic yoked stream did not change the registered outcome.

The current implementation therefore does not identify the paper's strongest action-feedback claim. It implements a closed interface, but the task can be solved without causally using the focal returned body stream in the required way.

11.6.2. Closed versus open: unresolved

Full minus open-loop transfer was favorable at 0.0547 but uncertain, with an interval spanning -0.0781 to 0.1875. Eight development seeds do not resolve this contrast, and the task offers a direct phase signal that reduces dependence on learned return.

11.6.3. Slow learning: unresolved

Full minus no-slow delayed transfer was 0.0625 with a wide interval. The full condition's mean slow-to-fast episode-step ratio was 0.0339, below the frozen 0.20 maximum, and slow updates survived fast reset. Those engineering checks verify two timescales, but they do not show that slow change caused the endpoint.

11.6.4. PWD prediction: conjunction fail

PWD improved return-prediction skill descriptively, but the interval and Holm gates failed. Moreover, the generic recurrent comparator had slightly higher Brier skill than the full loop. PWD use is identified for partial-cue action selection under P4, not as a superior general return-prediction representation.

11.7. Strongest alternative explanations

The pilot is compatible with a simpler interpretation: the synthetic event-phase tag contains a target-direction cue, route compatibility converts that cue into action preference, and exact mapping restores the correct action table after disruption. This explains P3 and P4 without requiring the agent to learn itself through contingent return.

Other retained explanations are:

1. the slow update is too weak or too nonspecific to affect transfer;
2. yoked body-return values enter the fast state but not the dominant route score;
3. the environment changes across episodes, limiting accumulation of mapping-specific advantage;
4. the generic comparator is weak on action selection yet competitive on return prediction; and
5. the oracle is target-aware but not obstacle-planning, so it is not a strict perfect ceiling.

11.8. Robustness requirements for Development Study 2

A new, pre-outcome development generator is justified only to test the diagnosed failure, not to retune the eight exposed seeds. It must:

1. remove direct target direction from phase tags;
2. make the action-to-return mapping latent and learnable only through contingent interaction;
3. keep yoked marginal return and phase statistics matched;
4. add a capacity-matched nonlinear recurrent or tabular expected-value control;
5. separate body-return dependence from route mapping;
6. increase independent development seeds without opening the final partition;
7. freeze generator, endpoints, and analysis before results;
8. retain the exact Development Study 1 null and all six contrasts; and
9. accept another null or adverse result without further seed-specific tuning.

11.9. Development Study 1 verdict

The controlled persistent-agent pilot is complete, reproducible, and scientifically mixed. Two bounded synthetic mechanism edges-mapping-specific rescue and receiver-assigned phase-route use-are supported under the frozen pilot. Four other primary claims fail or remain unresolved, including an exact null for closed contingent return versus yoked replay. The complete learning-loop mechanism is not identified. Development Study 2 may test the diagnosed shortcut under a separately frozen robustness generator; it may not rewrite Development Study 1 or present synthetic evidence as biological validation.

12. Development Study 2: shortcut-resistant robustness



Figure 5. Frozen paired contrasts from Development Study 2; five of seven gates pass.

Shortcut-Resistant Condition Outcomes

Means over sixteen seeds; favorable, null, and adverse comparator outcomes retained

	Evaluation	Transfer	Partial cue	Recovery	Brier skill
FULL LOOP	0.484	0.484	0.484	0.172	0.921
FIXED CONTROLLER	0.297	0.297	0.297	0.297	0.790
OPEN LOOP	0.250	0.250	0.250	0.250	0.812
YOKED REPLAY	0.250	0.250	0.250	0.219	0.776
PHASE SCRAMBLED	0.500	0.500	0.500	0.141	0.919
NO PWD	0.217	0.219	0.219	0.242	0.797
NO SLOW	0.250	0.250	0.250	0.250	0.812
WRONG MAP RESCUE	0.484	0.484	0.484	0.172	0.921
EXACT MAP RESCUE	0.484	0.484	0.484	0.656	0.921
GENERIC RECURRENT	0.484	0.484	0.484	0.172	0.921
LOOKUP ORACLE	1.000	1.000	1.000	1.000	0.466

Descriptive display only; inferential decisions use the frozen paired contrasts.

Figure 6. Descriptive condition means from Development Study 2, including the exact generic tie.

12.1. Why a second generator was necessary

Development Study 1 found an exact null between the full loop and yoked replay. Its event-phase tag also exposed target direction strongly enough that route compatibility could dominate the body-return path. That result could not establish that contingent action consequence caused later reconstruction. Development Study 2 did not rewrite or suppress the result. It created a new, independently seeded development generator designed around that diagnosed shortcut.

For seed (s), a hidden permutation

$$\pi_s : \{0, 1, 2, 3\} \rightarrow \{0, 1, 2, 3\} \quad (36)$$

maps cue identity to rewarded action. The phase feature for cue (c) is

$$\phi_i(c) = \begin{cases} 0.6, & i = c, \\ -0.1, & i \neq c. \end{cases} \quad (37)$$

Thus phase identifies a receiver-relative cue, not the rewarded action $\pi_s(c)$. The action mapping must be learned separately. During partial-cue episodes the direct cue is masked while the phase vector remains available.

12.2. Frozen design and provenance

The study crossed 16 development seeds with the same eleven named conditions used in the application interface. Each run contained 160 episodes: 80 training, 32 evaluation, 16 transfer after fast-state reset, 16 partial-cue probes, and 16 mapping-disruption/rescue episodes. The independent unit was the seed. All seven contrasts used complete seed pairs, 5,000 seed-cluster bootstrap resamples, exact paired sign flips, and one Holm family.

The contract, implementation, runner, and dedicated test file were frozen before any Development Study 2 result was generated. The complete test suite passed 64/64 at that boundary. The held-out final partition remained ungenerated and unopened.

The learned action value is updated only when the condition permits it:

$$Q_{t+1}(c, a) = 0.75Q_t(c, a) + 0.25r_t. \quad (38)$$

Yoked replay substitutes a deterministic nonfocal reward ($r_{\text{t}}^{\text{yoked}}$) for the action-contingent reward while preserving the same interface and episode schedule. Exact rescue rotates the learned action columns by the true disruption; wrong rescue applies a different rotation. No-slow, open-loop, fixed, no-PWD, phase-scrambled, oracle, and generic recurrent conditions remain explicit rather than being hidden inside one composite score.

12.3. Execution and deterministic replay

All 176 seed-condition runs completed. The canonical application produced 28,160 episode rows and no recorded failures. Primary and replay outputs were byte-identical:

Artifact	SHA-256
Run summaries	D68220B95F99C2D5A72DD4CA57208868F11C85237EFEB8C219A92D762A96D996
Episode results	1EF3B40E31176E7B310B809CC02FB3CBB6CE971A17E9051BB793AC8838906C4D
Primary contrasts	E91503D6DF3FECB4C201F3A5DE68521B0884BD67C31B9218672DAE02832BC048
Compact receipt	366370C2A2A1742B07D5CA684B432879EBD3F158D1FE72BCA722CD90F34AFB53

The summary, episode, and contrast digests are respectively

9feb6fefb690550431471dcaa3b021d3e816bada44593d63ac8eb6b534008279,
8088372038c23c2a263436b4faaec0e297ba6d74a6bbf9f32166618f62c0b169, and
5b8dae9251843810bf3bd6143c53174e3db335220384a9fcf24fdf17107243b4.

12.4. Preregistered results

For contrast (j), the estimator is

$$\hat{\tau}_j = \frac{1}{16} \sum_{s=1}^{16} (Y_{sj}^{\text{left}} - Y_{sj}^{\text{right}}). \quad (39)$$

A gate passes only when the lower 95% bootstrap bound exceeds its frozen practical floor and the Holm-adjusted p-value is below 0.05.

ID	Frozen contrast	Mean difference	95% interval	Holm p	Verdict
P1	Full minus yoked, retained improvement	0.2344	[0.1563, 0.3281]	0.00244	Pass
P2	Full minus open, transfer success	0.2344	[0.1406, 0.3281]	0.00391	Pass
P3	Exact minus wrong rescue, recovery	0.4844	[0.3906, 0.5781]	0.000214	Pass
P4	Full minus phase-scrambled, partial-cue success	-0.0156	[-0.1719, 0.1406]	1.00000	Null/unresolved
P5	Full minus no-slow, delayed transfer	0.2344	[0.1406, 0.3281]	0.00391	Pass
P6	Full minus no-PWD, calibrated return prediction	0.1239	[0.0928, 0.1501]	0.000366	Pass
P7	Full minus generic recurrent, evaluation success	0.0000	[0.0000, 0.0000]	1.00000	Exact null

The conjunctive result is

$$\sum_{j=1}^7 \mathbf{1}(G_j = 1) = 5, \quad G_{\text{complete}} = 0. \quad (40)$$

Five passing contrasts are not a complete pass. Two preregistered failures remain constitutive parts of the result.

12.5. What the favorable contrasts identify

12.5.1. Contingent return under the revised generator

Full exceeded yoked replay on retained mapping-specific improvement by 0.2344. Unlike Development Study 1, the outcome cannot be explained by a phase tag that directly names the rewarded action: the cue-to-action permutation varies by seed and is absent from the phase feature. Within this finite implementation, action-contingent return contributed to later performance.

This result does not establish subjective awareness, biological proprioception, or a privileged neural implementation. It establishes a causal edge inside the frozen synthetic generator.

12.5.2. Transfer and slow state

Full exceeded open loop and no-slow on transfer by the same 0.2344 mean difference. Fast state was explicitly reset before transfer, while the learned value table persisted. The result identifies a cross-episode learned state in this implementation. It does not identify synaptic plasticity, dendritic morphology, or a complete biological memory substrate.

12.5.3. Mapping-specific rescue

Exact remapping exceeded wrong remapping by 0.4844. The intervention changes which learned action column is reinstated after the known rotation. It therefore identifies mapping specificity, not a generic benefit from adding state or issuing a rescue label.

12.5.4. PWD input and prediction

The full loop exceeded no-PWD on evaluation Brier skill by 0.1239. In this task the phase vector carries cue/receiver identity and supports a better return prediction after the hidden action mapping is learned. This supports use of the implemented receiver-relative signal. It does not prove that biological PWDs use the same code or that oscillatory phase is uniquely required for prediction.

12.6. The two boundaries that prevent overclaiming

12.6.1. Phase scrambling did not impair partial-cue success

P4 failed. Phase scrambling changed the decoded receiver but did not reliably lower partial-cue success across the balanced seed permutations. The phase-scrambled condition's descriptive partial-cue mean was 0.500, compared with 0.484 for the full loop. This reveals an important symmetry: across the chosen permutations, a consistent receiver shift can sometimes be relearned rather than functioning as destructive incoherence.

The result prevents the manuscript from claiming that any mismatch in receiver-relative phase assignment necessarily destroys reconstruction. A stronger biological or artificial test must distinguish arbitrary relabeling that can be relearned from transient or context-dependent incompatibility that cannot.

12.6.2. The generic recurrent learner tied the full loop exactly

The full and generic recurrent conditions used the same phase-derived cue and the same tabular update budget. Their evaluation success was identical for every seed, producing a mean difference and confidence interval of exactly zero. This is not a defect to hide. It is the strongest nonuniqueness result in the study.

The implementation demonstrates a learning-loop operator, not a SAN-exclusive algorithm. The SAN vocabulary supplies an interpretable decomposition-receiver identity, phase compatibility, returned consequence, persistent disposition, and reinstatement—but the frozen result does not show superior prediction over an equivalently informed generic recurrent learner.

12.7. Relation to the action-feedback cycle, proprioception, and GCS

The synthetic episodes instantiate only a minimal functional analogue of the broader SAN action-feedback cycle. A cue and receiver-relative phase pattern initiate an internal action selection; action changes the reward-bearing relation; returned consequence changes a persistent disposition; and that disposition affects the next reconstruction. The loop therefore tests the logical dependency of action, return, memory, and later state without pretending to reproduce cortical anatomy.

In the biological proposal, proprioceptive and interoceptive return are not decorative confirmations after action. They are evidence about what the body did, what changed, and whether the selected organization succeeded. The corrected Gamma Consideration Sandwich places primary-sensory-side alpha and prefrontal beta input at the Layer 4 intake side, Layer 2/3 gamma consideration and proprioceptive coordination in the functional middle, and Layer 5 theta/gamma action-side output plus returned body input at the other side. Development Study 2 operationalizes the closure distinction, not those frequency-layer assignments themselves.

12.8. Artificial Neurology interpretation

For Artificial Neurology, the result supports a typed engineering pattern:

$$\text{receiver-relative input} \rightarrow \text{state-conditioned action} \rightarrow \text{returned consequence} \rightarrow \text{permissioned persistent update} \rightarrow \text{later reconstruction.} \quad (41)$$

The pattern is auditable because each transition carries a seed, condition, episode, phase, state-parent hash, result hash, and event hash. Permission boundaries distinguish a controller that can update persistent state from fixed, open, and no-slow alternatives. Exact rollback and mapping rescue are tested separately.

The generic tie adds an equally important engineering lesson: naming a component after a biological or SAN concept does not establish its necessity. Competing architectures must receive the same usable observations, update budget, and evaluation partition. If a generic recurrent implementation realizes the same operator, the honest claim is functional equivalence under the tested conditions.

12.9. Development Study 2 verdict

Development Study 2 repairs the specific direct-action shortcut diagnosed in Development Study 1 and produces reproducible evidence for contingent return, persistent transfer, exact remapping, and use of a receiver-relative phase signal in a finite synthetic generator. It also produces two decisive limitations: phase scrambling is not harmful in the registered partial-cue test, and a matched generic recurrent learner ties the full loop exactly.

The evidence therefore advances the paper as a bounded mechanism-and-methods paper while rejecting uniqueness and biological-validation claims. The next gate is formal verification of finite interface, provenance, permutation, update, gate-accounting, and adverse-history invariants. A later review must reconstruct the strongest version of the thesis before challenging it, preserve all Development Study 1 and Development Study 2 nulls, and keep the paper distinct from the published work on explicit network self-knowledge.

13. Bounded formal verification

13.1. Formal object and scope

The Lean model encodes four cue identities, four action identities, eleven conditions, five phases, update permissions, an abstract bijective cue-action mapping, a one-step action rotation with inverse, seven result verdicts, a compact custody receipt, and four claim classes. It is intentionally smaller than the executable application. The compatibility ledger distinguishes exact encodings, partial encodings, abstract interfaces, and result-ledger imports.

The central boundary can be written as

$$\mathcal{F}_{D8} \vdash C_{finite} \not\Rightarrow C_{biology} \vee C_{consciousness} \vee C_{uniqueness}. \quad (42)$$

Here $\mathcal{F}_{D8}$ is the formal finite model. A proof about it is a proof about that model, not an empirical bridge theorem.

13.2. Interface and mapping theorems

The compiler verifies the declared cardinalities and that the receiver tag is a function of cue identity without an action argument. It proves that direct cue input is unavailable in the partial-cue phase and available during evaluation. The update-permission table permits updates for the full and yoked conditions while refusing them for fixed, open-loop, no-slow, and oracle conditions.

An abstract mapping contains forward and inverse functions plus left- and right-inverse obligations. Lean proves both round trips. A concrete four-action rotation and its inverse also round trip for every action. These are interface and algebraic invariants. They do not prove that every seeded Python mapping is represented unless the separately tested compatibility edge is satisfied.

13.3. Result and adverse-history theorems

The sealed Development Study 2 verdict vector contains seven entries:

$$(1, 1, 1, 0, 1, 1, 0). \quad (43)$$

Lean verifies its length, its exact pass count of five, and the failure of the all-pass gate. It separately encodes the full-minus-generic numerator as zero and proves that zero cannot clear a positive natural-number floor. The phase-scrambling numerator is encoded as (-1) over 64 and proved nonpositive.

The adverse-history theorem is generic:

$$\text{take}_{|H|}(H \mathrel{++} N) = H. \quad (44)$$

Appending a later study (N) preserves the earlier ledger prefix (H). This does not make earlier interpretation immutable; it prevents later favorable evidence from silently deleting earlier nulls and failures. Contextual reinterpretation must still reconstruct the original strongest-form claim before revising it.

13.4. Receipt and claim-class theorems

The compact receipt encodes a frozen contract, four scientific output files, byte-identical replay, zero recorded run failures, and an ungenerated held-out partition. Lean proves those encoded values. Their link to disk bytes remains governed by SHA-256 manifests and replay checks, not by the theorem alone.

The claim classifier authorizes finite synthetic claims and refuses biological, consciousness, and uniqueness classes. These refusal theorems make the evidence boundary executable:

$$A(c) = \begin{cases} 1, & c = \text{finite synthetic}, \\ 0, & c \in \{\text{biological, consciousness, uniqueness}\}. \end{cases} \quad (45)$$

13.5. Compiler and axiom audit

The accepted r3 invocation used the shared compiler lease and exited 0. All 33 declarations were followed by `#print axioms`. Eighteen report no dependencies; fifteen report only standard `propext`. No declaration reports `sorryAx`, `Classical.choice`, or a custom axiom.

Two earlier attempts remain visible. The first was stopped after an unnecessarily broad Mathlib import caused dependency loading beyond the wrapper's outer timeout. The second reached Lean and failed on one unavailable order-lemma name. The repaired third attempt changed the zero numerator and positive floor to natural-number count-scale quantities and proved the same boundary by simplification. Neither failed attempt is reported as a theorem pass.

13.6. What formal verification changes

Formal verification makes several errors harder:

1. misreporting five passes as seven;
2. calling the complete gate successful;
3. claiming that an exact zero clears a positive superiority floor;
4. granting persistent updates to a frozen no-slow or open-loop condition;
5. treating an ungenerated held-out partition as observed;
6. deleting earlier adverse history when a later study is appended; and
7. promoting finite synthetic results into biological, consciousness, or uniqueness claims.

It does not repair the failed P4 or P7 contrasts, prove the bootstrap or sign-flip implementation, or provide an external application. The Python tests, result replay, source ledger, scientific literature, and future empirical work retain separate evidential roles.

13.7. The formal gate verdict

The bounded formal gate passes. The formal model is useful because it proves finite structural and accounting claims while explicitly refusing the larger inferences. The release review must now examine the full paper rather than the formal leaf alone: source genealogy, scientific alignment, strongest-form thesis, generic nonuniqueness, phase-scrambling null, neighboring-paper ownership, figures, citations, and release language all require independent review before a clean final preprint can be assembled.

14. Cumulative evidence and claim boundary

The evidence layers answer different questions. Source recovery establishes what the historical proposal said and when each extension entered it. Component neuroscience establishes that prediction, typed body return, mismatch, phase-sensitive interaction, partial-cue recall, and coadaptation are real operations, while leaving the joined SAN architecture open. The causal model states which interventions could identify individual edges. The executable system shows that the declared interfaces and controls can be built. The development studies test those interfaces under frozen finite generators. Lean verifies finite accounting and scope invariants.

The cumulative result is mixed rather than contradictory. Development Study 1 shows that a phase-labeled route can solve part of a task without using the focal contingent return in the required way. Development Study 2 removes that shortcut and identifies a contingent-return effect, but also shows that stable receiver relabeling can be relearned and that a generic recurrent learner can implement the same finite operator. The strongest currently earned statement is therefore functional and nonexclusive: a typed action-return-update-reconstruction loop can be implemented, causally decomposed, replayed, and formally bounded.

The following claims remain unavailable: that the synthetic agent is conscious; that the complete GCS anatomy is established; that biological PWD uses the implemented code; that phase scrambling generally impairs reconstruction; that SAN is uniquely necessary for the operator; or that a held-out, human, animal, or clinical result exists. The final held-out partition was not generated or opened.

15. Alternative explanations and retirement conditions

The added mechanism is unsupported in a tested setting if any of the following fully accounts for the result:

1. firing rate, power, familiarity, salience, arousal, movement, reward, or task history;
2. a generic recurrent world model under matched resources;
3. noncontingent replay with the same temporal and marginal statistics;
4. explicit report strategy or generic agency cues;
5. arbitrary rather than mapping-specific rescue;
6. hidden common input that creates apparent coupling without directed effect;
7. train-test leakage or architecture-specific privileged information;
8. post-treatment adjustment that blocks or creates the measured effect; or
9. an estimator that labels any structured signal as PWD without receiver-relative prediction.

The phase-learning extension must be narrowed in a tested domain if receiver-relative timing adds no predictive or causal value after the controls, if the PWD estimator cannot outperform simpler rate or residual features, or if incompatible internal routes yield the same claimed mechanism label. Such a result identifies an unearned edge or implementation; it does not erase the source genealogy or the broader fact of embodied learning.

16. Relationship to neighboring SAN papers

The current 103-paper public corpus contains multiple ancestors and neighboring mechanisms. The present paper cites rather than absorbs them.

Neighbor	Existing ownership	Residual owned here
SAN theory-and-methods framework [47]	Broad phase-gradient routing and reconstruction program	Focused whole-loop mechanism and direct controls
Field-Cell Self [48]	Observer identity hypothesis	Learned action-return dynamics
Embodied Neural Rendering [49]	Full GCS/body-map treatment	GCS as one junction inside loop learning
Coherence-Dispersion [50]	Coupling and differentiation operator	Its role in later self-world reconstruction
Emotional State Changes [51]	Emotion-specific self-registration	General perception-action learning loop
Network Learns It Is a Network [52]	Explicit declarative network self-knowledge	Online self-learning before explicit explanation
External Cortex Loop [53]	External contingency incorporation	Internal embodied loop as general architecture
Building Sentient Beings [54]	Joint developmental and sentence architecture	Narrow prerequisite; joint attribution preserved
Gamma to Informative Difference [55]	Shannon-PWD operator	Integration into action-return learning
Multiscale Causation [56]	Cross-scale causal taxonomy	Temporal closure and reconstruction
Dialogue as a Coupled Neural System [57]	Interpersonal coupling and dialogue	Individual embodied whole-loop learning

No reviewed neighboring paper centrally owns the exact joined scope: closed embodied learning loop, whole-loop self, receiver-relative PWD, phase-compatible recurrent recruitment, differentiated self-world reinstatement, and action-return plasticity.

17. Discussion

The phrase "a learning loop that learns itself" is compact enough to invite opposite errors. One reduces it to a symbol that refers to itself. The other inflates it into a complete explicit copy of the system. The source context supports neither. The self is the continuing embodied process whose operation repeatedly helps generate the evidence from which it learns; self-models are contents and constraints within that larger process.

Hofstadter's feedback examples explain how recursion can stabilize a pattern without supplying SAN's neural mechanism. SAN's phase extension begins where the analogy ends: recurrence occurs through learned tissue whose effective interaction depends on anatomy, receiver readiness, relative timing, inhibition, and context. Resonance here does not mean perfect global synchrony. It means preferential recruitment of compatible learned routes, with PWD and structured dispersion preventing content from collapsing into uniform coordination.

Proprioception makes the theory genuinely embodied. The loop is constrained by a body it moves and through which it senses. GCS proposes a more specific functional junction: primary-sensory-side alpha and prefrontal beta inputs at the Layer 4 intake side, Layer 2/3 gamma consideration and proprioceptive coordination, and Layer 5 theta/gamma action-side output plus returned body input. Current component studies do not establish that complete mapping. Its present scientific value is to force exact source, layer, direction, timing, and consequence tests.

The application results sharpen rather than simply confirm the theory. The first generator exposed a shortcut and failed to identify contingent return. The second identified contingent return after removing rewarded action from the phase tag, yet its phase-scrambling null and generic tie limit the phase and uniqueness claims. SAN currently contributes an interpretable decomposition and falsification program, not an exclusive algorithm.

Formal verification serves the same discipline. It proves that the finite package contains five passes rather than seven, that an exact zero cannot clear a positive floor, that later evidence preserves earlier adverse history, and that the declared result class does not authorize biology, consciousness, or uniqueness. It does not replace empirical work.

The next scientific step is not more tuning on these exposed generators. It is a separately frozen external study with an independently implemented strong comparator, architecture-blind evaluation, real or biologically constrained typed return, transient and context-dependent phase incompatibility rather than a relearnable stable permutation, and an explicit mediation plan. Until then, the present work is a bounded synthetic theory-and-methods preprint.

18. Conclusion

A loop can refer to itself without learning itself, and a system can learn a self-model without identifying the process that continuously produces the self. SAN's learning-loop formulation joins prediction, embodiment, action, returned evidence, receiver history, and later self-world reconstruction within one continuing process.

The paper establishes a source-faithful genealogy, a biologically bounded synthesis, an explicit causal and mathematical contract, a working typed application, two frozen development studies, exact deterministic replay, and a 33-theorem Lean boundary layer. It also establishes what the current evidence does not show. The first study contains an exact closed-versus-yoked null. The second fails its phase-scrambling gate and ties a matched generic recurrent learner exactly. The complete mechanism and uniqueness gates therefore remain open.

The earned contribution is a reproducible and falsifiable decomposition of the learning loop that learns itself. It provides a disciplined route from recursive self-reference to embodied learning and receiver-relative reinstatement without treating synthetic performance, formal verification, or plausible self-description as proof of consciousness.

References

1. Hofstadter, D. R. *Gödel, Escher, Bach: An Eternal Golden Braid*. Basic Books (1979).
2. Hofstadter, D. R. *I Am a Strange Loop*. Basic Books (2007).
3. Hawkins, J. & Blakeslee, S. *On Intelligence*. Times Books (2004).
4. Shannon, C. E. A mathematical theory of communication. *Bell System Technical Journal* **27**, 379-423, 623-656 (1948). <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
5. Rao, R. P. N. & Ballard, D. H. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience* **2**, 79-87 (1999). <https://doi.org/10.1038/4580>

6. Botvinick, M. & Cohen, J. Rubber hands “feel” touch that eyes see. *Nature* **391**, 756 (1998). <https://doi.org/10.1038/35784>
7. Blanke, O., Ortigue, S., Landis, T. & Seeck, M. Stimulating illusory own-body perceptions. *Nature* **419**, 269-270 (2002). <https://doi.org/10.1038/419269a>
8. Critchley, H. D., Wiens, S., Rotshtein, P., Öhman, A. & Dolan, R. J. Neural systems supporting interoceptive awareness. *Nature Neuroscience* **7**, 189-195 (2004). <https://doi.org/10.1038/nm1176>
9. Sommer, M. A. & Wurtz, R. H. A pathway in primate brain for internal monitoring of movements. *Science* **296**, 1480-1482 (2002). <https://doi.org/10.1126/science.1069590>
10. Keller, G. B., Bonhoeffer, T. & Hübener, M. Sensorimotor mismatch signals in primary visual cortex of the behaving mouse. *Neuron* **74**, 809-815 (2012). <https://doi.org/10.1016/j.neuron.2012.03.040>
11. Saleem, A. B., Ayaz, A., Jeffery, K. J., Harris, K. D. & Carandini, M. Integration of visual motion and locomotion in mouse visual cortex. *Nature Neuroscience* **16**, 1864-1869 (2013). <https://doi.org/10.1038/nm.3567>
12. Schneider, D. M., Nelson, A. & Mooney, R. A synaptic and circuit basis for corollary discharge in the auditory cortex. *Nature* **513**, 189-194 (2014). <https://doi.org/10.1038/nature13724>
13. Azim, E., Jiang, J., Alstermark, B. & Jessell, T. M. Skilled reaching relies on a V2a propriospinal internal copy circuit. *Nature* **508**, 357-363 (2014). <https://doi.org/10.1038/nature13021>
14. Fink, A. J. P. et al. Presynaptic inhibition of spinal sensory feedback ensures smooth movement. *Nature* **509**, 43-48 (2014). <https://doi.org/10.1038/nature13276>
15. Attinger, A., Wang, B. & Keller, G. B. Visuomotor coupling shapes the functional development of mouse visual cortex. *Cell* **169**, 1291-1302.e14 (2017). <https://doi.org/10.1016/j.cell.2017.05.023>
16. Leinweber, M., Ward, D. R., Sobczak, J. M., Attinger, A. & Keller, G. B. A sensorimotor circuit in mouse cortex for visual flow predictions. *Neuron* **95**, 1420-1432.e5 (2017). <https://doi.org/10.1016/j.neuron.2017.08.036>
17. Schneider, D. M., Sundararajan, J. & Mooney, R. A cortical filter that learns to suppress the acoustic consequences of movement. *Nature* **561**, 391-395 (2018). <https://doi.org/10.1038/s41586-018-0520-5>
18. London, B. M. & Miller, L. E. Responses of somatosensory area 2 neurons to actively and passively generated limb movements. *Journal of Neurophysiology* **109**, 1505-1513 (2013). <https://doi.org/10.1152/jn.00372.2012>
19. Zobeiri, O. A. & Cullen, K. E. Cerebellar Purkinje cells combine sensory and motor information to predict the sensory consequences of active self-motion in macaques. *Nature Communications* **15**, 4003 (2024). <https://doi.org/10.1038/s41467-024-48376-0>
20. Cross, K. P., Cook, D. J. & Scott, S. H. Rapid online corrections for proprioceptive and visual perturbations recruit similar circuits in primary motor cortex. *eNeuro* **11**, ENEURO.0083-23.2024 (2024). <https://doi.org/10.1523/ENEURO.0083-23.2024>
21. Womelsdorf, T. et al. Modulation of neuronal interactions through neuronal synchronization. *Science* **316**, 1609-1612 (2007). <https://doi.org/10.1126/science.1139597>
22. Bosman, C. A. et al. Attentional stimulus selection through selective synchronization between monkey visual areas. *Neuron* **75**, 875-888 (2012). <https://doi.org/10.1016/j.neuron.2012.06.037>
23. van Kerkoerle, T. et al. Alpha and gamma oscillations characterize feedback and feedforward processing in monkey visual cortex. *Proceedings of the National Academy of Sciences USA* **111**, 14332-14341 (2014). <https://doi.org/10.1073/pnas.1402773111>
24. Bastos, A. M. et al. Visual areas exert feedforward and feedback influences through distinct frequency channels. *Neuron* **85**, 390-401 (2015). <https://doi.org/10.1016/j.neuron.2014.12.018>
25. Montemurro, M. A., Rasch, M. J., Murayama, Y., Logothetis, N. K. & Panzeri, S. Phase-of-firing coding of natural visual stimuli in primary visual cortex. *Current Biology* **18**, 375-380 (2008). <https://doi.org/10.1016/j.cub.2008.02.023>
26. Tort, A. B. L., Komorowski, R. W., Manns, J. R., Kopell, N. J. & Eichenbaum, H. Theta-gamma coupling increases during the learning of item-context associations. *Proceedings of the National Academy of Sciences USA* **106**, 20942-20947 (2009). <https://doi.org/10.1073/pnas.0911331106>
27. Rodriguez, E. et al. Perception's shadow: long-distance synchronization of human brain activity. *Nature* **397**, 430-433 (1999). <https://doi.org/10.1038/17120>
28. Melloni, L. et al. Synchronization of neural activity across cortical areas correlates with conscious perception. *Journal of Neuroscience* **27**, 2858-2865 (2007). <https://doi.org/10.1523/JNEUROSCI.4623-06.2007>

29. Pitts, M. A., Padwal, J., Fennelly, D., Martínez, A. & Hillyard, S. A. Gamma band activity and the P3 reflect post-perceptual processes, not visual awareness. *NeuroImage* **101**, 337-350 (2014). <https://doi.org/10.1016/j.neuroimage.2014.07.024>
30. Nakazawa, K. et al. Requirement for hippocampal CA3 NMDA receptors in associative memory recall. *Science* **297**, 211-218 (2002). <https://doi.org/10.1126/science.1071795>
31. Liu, X. et al. Optogenetic stimulation of a hippocampal engram activates fear memory recall. *Nature* **484**, 381-385 (2012). <https://doi.org/10.1038/nature11028>
32. Ganguly, K. & Carmena, J. M. Emergence of a stable cortical map for neuroprosthetic control. *PLOS Biology* **7**, e1000153 (2009). <https://doi.org/10.1371/journal.pbio.1000153>
33. Sadtler, P. T. et al. Neural constraints on learning. *Nature* **512**, 423-426 (2014). <https://doi.org/10.1038/nature13665>
34. Orsborn, A. L. et al. Closed-loop decoder adaptation shapes neural plasticity for skillful neuroprosthetic control. *Neuron* **82**, 1380-1393 (2014). <https://doi.org/10.1016/j.neuron.2014.04.048>
35. Oby, E. R. et al. New neural activity patterns emerge with long-term learning. *Proceedings of the National Academy of Sciences USA* **116**, 15210-15215 (2019). <https://doi.org/10.1073/pnas.1820296116>
36. Hafner, D., Pasukonis, J., Ba, J. & Lillicrap, T. Mastering diverse control tasks through world models. *Nature* **640**, 647-653 (2025). <https://doi.org/10.1038/s41586-025-08744-2>
37. Turrigiano, G. G., Leslie, K. R., Desai, N. S., Rutherford, L. C. & Nelson, S. B. Activity-dependent scaling of quantal amplitude in neocortical neurons. *Nature* **391**, 892-896 (1998). <https://doi.org/10.1038/36103>
38. Blumberg, M. September 21, 2011 source note: “The learning loop that learns itself.” Publicly Git-fixed June 8, 2022. https://github.com/v5ma/selfawarenetworks/blob/7d1431213cd4ac9788848a5396ec2992ece11cfd/a0051z_ctpr.txt
39. Blumberg, M. Pre-September 21, 2011 dialogue excerpts: senses, brainwaves, actions, and the whole learning loop. Publicly Git-fixed June 8, 2022. https://github.com/v5ma/selfawarenetworks/blob/7d1431213cd4ac9788848a5396ec2992ece11cfd/a0072z_ctpr.txt
40. Blumberg, M. April 19, 2012 source note: prediction and the cellular player-listener problem. Publicly Git-fixed June 8, 2022. https://github.com/v5ma/selfawarenetworks/blob/5360f6414909a2474129b2140ee0dc3112c93b6b/a0632z_ctpr.txt
41. Blumberg, M. September 18, 2012 source note: learning loops inside learning loops. Publicly Git-fixed June 8, 2022. https://github.com/v5ma/selfawarenetworks/blob/37da575ba33f85cef4239295be161066d8d39f4c/a0119z_ctpr.txt
42. Blumberg, M. Undated SAN note: learning playback, phase coincidence, inhibition, decoherence, and learned routes. Publicly Git-fixed June 8, 2022. https://github.com/v5ma/selfawarenetworks/blob/7d1431213cd4ac9788848a5396ec2992ece11cfd/a0049z_ctpr.txt
43. Blumberg, M. Artificial Neurology program map. Publicly Git-fixed June 8, 2022, commit 5360f6414909a2474129b2140ee0dc3112c93b6b.
44. Blumberg, M. *Artificial Neurology: Robotic State & Stages*. Publicly Git-fixed November 28, 2022, commit 4e0cf1ee0975c01dc57182368497d3f0cd864030.
45. Blumberg, M. *Consideration Sandwich: Gamma Waves in the Middle*. Author passages dated October 21, 2024, commit 7bbb9e317793480ab2a08781f8801ee224f37d44.
46. Blumberg, M. Gamma Consideration Sandwich role clarification. Author clarification, August 12, 2026; preserved in the project source-recovery receipt.
47. Blumberg, M. *Self-Aware Networks: A Falsifiable Theory-and-Methods Framework for Phase-Gradient Routing and Recurrent Oscillatory Reconstruction*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21693884>
48. Blumberg, M. *The Field-Cell Self: A Falsifiable Mechanistic Identity Hypothesis for Conscious Neural Rendering*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21693922>
49. Blumberg, M. *Embodied Neural Rendering in Self-Aware Networks*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21693949>
50. Blumberg, M. *The Coherence-Dispersion Sculpting Hypothesis*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21693957>
51. Blumberg, M. *How a Brain-Body Network Detects Its Own Emotional State Changes*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21693970>
52. Blumberg, M. *The Network That Learns It Is a Network: From Predictive Selfhood to Explicit Network Self-Knowledge*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21743941>
53. Blumberg, M. *The External Cortex Loop: Contingency Interruption, Reinstatement, and Cognitive Incorporation in EEG Neurofeedback*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21743944>

54. Blumberg, M. & Miller, M. *Building Sentient Beings*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21061762>
55. Blumberg, M. *From Gamma Synchrony to Informative Difference: A Shannon-Theoretic Model of Tonic Context and Phase-Wave Differentials*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21861046>
56. Blumberg, M. *Multiscale Causation in Self-Aware Networks: Bottom-Up Construction, Top-Down Constraint, and Lateral Coordination*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21925367>
57. Blumberg, M. *Dialogue as a Coupled Neural System: Interpersonal Synchronization, Shared Reference Frames, and Informative Divergence*. Zenodo (2026). <https://doi.org/10.5281/zenodo.21926103>

Declarations

Preprint status. This manuscript is a preprint and has not been peer reviewed.

Data and code availability. The public companion contains the frozen contracts, typed reference application, test suite, primary/replay outputs, figures, claim ledgers, formal source, compiler receipt, and exact manifests. The held-out final partition was not generated and is not included.

Competing interests. The author declares no competing financial interest.

Ethics. No human participants, animals, clinical records, or private biological data were used. All reported application results are deterministic synthetic development evidence.

Evidence boundary. The work does not establish biological consciousness, artificial sentience, personal identity, the complete GCS architecture, phase necessity, or SAN-specific superiority.